

Logan Chook

Electrical Engineering Student | Embedded Systems & Firmware

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EDUCATION

Bachelor of Science in Electrical Engineering

Focus: Embedded Systems, Hardware-Software Integration, Robotics

TECHNICAL SKILLS

Languages & Tools: C, C++, Python, CMake, Git, Linux, Bash

Embedded & Protocols: ESP32, QEMU, I2C, SPI, UART, CAN, Interrupts, FreeRTOS

Hardware & Robotics: Altium Designer, Flex PCB Design, ROS 2, MoveIt, RViz

PROJECTS

ESP32 Firmware Emulation in QEMU

- Built QEMU-compatible ESP32 firmware workflow with CMake build support
- Mocked pedal-sensor and peripheral layer for fault injection without physical vehicle

Robotic Arm Research

- Conducted research utilizing ROS 2, MoveIt, and RViz for simulated-to-real robot motion

Interrupt-Driven IMU Firmware

- Designed non-blocking I2C IMU firmware utilizing hardware interrupts & static memory

Battery Temperature Flex PCB

- Designed compact flex-PCB for battery temperature monitoring in Altium Designer